MIL-P-37564
8 February 1977

MILITARY SPECIFICATION

PLUGS, EAR, HEARING PROTECTION, UNIVERSAL SIZE,

SINGLE FLANGE WITH HEADBAND

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of hearing protection devices which provide a positive seal by distributing canal cap pressure over the ear canal for use in environments in which high intensity sound is encountered.

1.2 Classification. Ear plugs shall be furnished in the following types and as specified in 6.1.

Types

I (without insulation on headband).
II (with insulation on headband).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-B-585 - Box, Wood, Wirebound.
PPP-B-601 - Box, Wood, Cleated-Plywood.
PPP-B-621 - Box, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Shipping Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 S. 20th St., Phila., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:00Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL

FED-STD-601 - Rubber, Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC) Drawing

No. 23220 PLUG, EAR, HEARING PROTECTION WITH HEAD BAND.
No. 23306 PLUG, EAR, HEARING PROTECTION, UNIVERSAL SIZE, PLASTIC.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

S3.19 Method for the Measurement of the Real-Air Attenuation of Ear Protectors at Threshold.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Canal caps (Type I). The canal caps shall be made of neoprene and have a Shore Durometer A hardness value of 9 ± 1 when tested as described in 4.4.1. The canal caps shall be non-allergenic and resistant to body oils and acids.

Source: https://assist.dla.mil -- Downloaded: 202640-21T11:00Z
Check the source to verify that this is the current version before use.

3.1.2 Canal caps (Pods, Type II). The canal caps (pods) shall be made of vinyl plastic and have a Shore Durometer A hardness value of 24 + 2 when tested as described in 4.4.1. The canal caps (pods) shall be non-allergenic and resistant to body oils and acids.

3.1.3 Retainers. For the Type I, the retainers shall be made of neoprene and have a Shore Durometer A hardness value of 35 + 2 when tested as described in 4.4.1. For the Type II, the screws shall be zinc plated steel and the nuts shall be nickel or cadmium plated brass.

3.1.4 Head band. The head band shall be made of austenitic stainless steel enclosed in a polyvinyl chloride tube. For the Type II, the head band polyvinyl chloride tube shall have a layer of polyurethane foam on the surface to be in contact with the wearer.

3.2 Design and construction. The ear plug shall consist of a canal cap or pod for each ear. Each canal cap (pod) shall fit into and seal the external auditory meatus. The canal caps are attached to the head band by the retainers. The Type I ear plug shall be in accordance with Defense Personnel Support Center Drawing No. 23220. The Type II ear plug shall be in accordance with DPSC Drawing No. 23306.

3.3 Sound attenuation. Sound attenuation shall be determined in accordance with the method described in 4.4.2. Attenuation shall exceed the following minimum values:

Frequency (Hz)	Type I Attenuation (db)	Type II Attenuation (db)
500	20	15
1000	26	18
2000	35	29
3000	37	36
4000	36	39
6000	--	35
8000	--	36

3.4 Workmanship. The finished ear plug shall conform to the highest standards of commercial fabrication. The ear plug shall be free from defects which detract from its appearance or may impair its serviceability. The ear plug shall not cause irritation and shall be free from objectionable odor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for

Source: https://assist.dla.mil -- Downloaded: 202640-21T11:00Z
Check the source to verify that this is the currentversion before use.

the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the rubber, plastic, and stainless steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in table I. Unit of product for examination purposes shall be one ear plug.

TABLE I. Sampling for examination.

Type of defect	Inspection level	AQL (percent defective)
Major	S-2	1.0
Minor	S-1	2.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:00Z
Check the source to verify that this is the current version before use.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in table II below. The unit of product for test purpose shall be one ear plug.

TABLE II. Sampling for tests.

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Hardness	3.1.1 and 3.1.2	4.4.1	S-1	1.0
Attenuation	3.3	4.4.2	S-1	1.0

4.2.3 Sample unit usage. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The ear plug shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in tables I and III. Examination, however, shall not be restricted to defects listed in table III. Any ear plug in the sample containing one or more defects shall be rejected and, if the number of defective ear plugs in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Head band does not hold adjustment.
102	Dimensions not in agreement with drawing.
103	Surfaces of ear caps in contact with wearer not free from roughness, burrs, or slivers.
Minor	
201	Surfaces not free of discoloration.
202	Feathers or burrs present on parts of ear plug other than parts in contact with wearer.

Source: https://assist.dla.mil -- Downloaded: 202640-21T11:00Z
Check the source to verify that this is the currentversion before use.

4.3.1 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Hardness of canal caps and retainers. The hardness of the ear caps and retainers or pods shall be determined in accordance with Method 3021 of Federal Test Method Standard No. 601.

4.4.2 Sound attenuation. Sound attenuation shall be determined in accordance with ANSI Standard S3.19.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each ear plug shall be packaged in a self-sealing plastic bag. Method of packaging shall be in accordance with MIL-P-116, method III.

5.1.1.2 Intermediate packaging. Ten (10) ear plugs of type I or twelve (12) ear plugs of type II shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure of box shall be as specified in the appendix of PPP-B-636.

5.1.2 Level C. Ear plugs shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. One hundred (100) ear plugs of type I or one hundred forty four (144) ear plugs of type II shall be packed in an exterior container designed for a type 2 load, and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2 or PPP-B-636, class weather resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

Source: https://assist.dla.mil -- Downloaded: 202610-21T11:00Z
Check the source to verify that this is the current version before use.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with the appendix of PPP-B-636.

5.2.2 Level B. One hundred (100) ear plugs of type I or one hundred forty four (144) ear plugs of type II shall be packed in an exterior container designed for a type 2 load, and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit, intermediate and exterior container. Each unit, intermediate and exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Selection of applicable levels of packaging and packing.
d. Applicable NSN.

6.2 NSN coverage. This specification covers the following items appearing in the Federal Supply Catalog:

National Stock No.	Item Identification	Type
6515-00-181-8058	PLUG, EAR, Hearing Protection, Universal Size, Single Flange, without Insulation on Headband	I
6515-00-392-0726	PLUG, EAR, Hearing Protection, Universal Size, Single Flange, with Insulation on Headband	II

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-2026

Source: https://assist.dla.mil -- Downloaded: 202640-21T11:00Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-P-37564 PLUGS, EAR, HEARING PROTECTION, UNIVERSAL SIZE, SINGLE FLANGE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202640-21T11:00Z
Check the source to verify that this is the current version before use.